Onshore energy services

Current landscape and challenges

Weathernews Inc.

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AGENDA

1. Our approach for data science projects
2. Wind & solar generation forecasts
3. Gas & Power demand forecasts
4. Specific use cases
5. Next steps

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1. Our approach for data science projects

WNI experiences

Our experience in other sectors outside of energy has enabled us to gain new knowledge

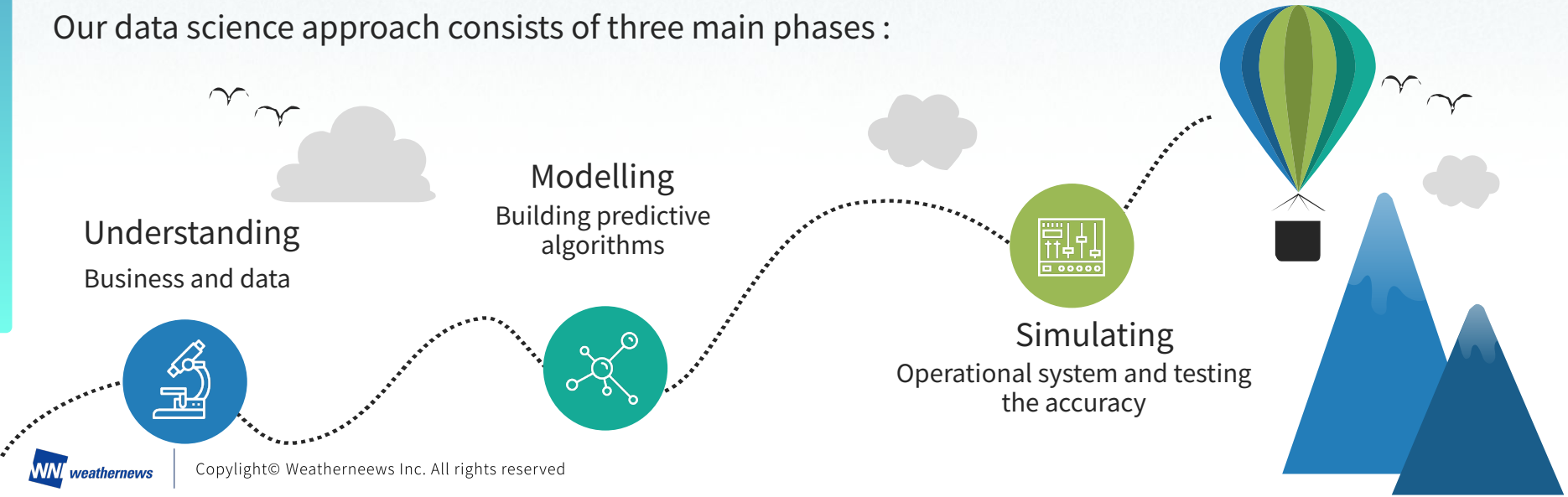
Multi-skilled team

Our teams based on a strong R&D foundation allows the generation of new ideas.

WNI approach

Through different projects, we developed our own work methodology for data-driven studies

Our data science approach consists of three main phases :



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1. Our approach for data science projects

1	2	3	4	5
Data Cleaning	**Business Data**	**Wx sources**	**Modelling**	**Deployment**
With automated scripts and / or ML algorithms	Analysis of all available data & Feature Engineering	Using wx data from traditional models & innovative models	Statistical modelling innovation & post-process improvements	Models deployment, industrialisation and scaling up

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2. Wind & solar generation forecasts

- Current situation:
  - Weather sources : ECMWF, ICON-EU, AROME…
  - Features : wind speed, solar radiation, temperature…
  - Algorithms : catboost, xgboost, bagging / stacking

- Current challenges :
  - Accuracy (mainly solar)
  - Productivity (data cleaning, retraining system, reporting...)

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3. Gas & Power demand forecasts

- Current situation :
  - Weather sources : In-house weather combination
  - Features : past business data (H-1, D-1…), calendar, temperature, wind speed…
  - Algorithm : MARS

- Current challenges :
  - Accuracy (mainly day-ahead forecasts)
  - Migration to new system

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4. Specific use cases

- Example of specific services : Power loss forecasts, “AI-Weather forecasts”, DLR...
- Current situation :
  - Weather sources : ECMWF, ICON-EU, AROME… + probabilistic sources
  - Features : temperature, other business data..
  - Algorithm : state-of-the-art algorithms

- Current challenges :
  - Productivity
  - Migration to more robust environment

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5. Next steps

- Q&A session

- Challenge proposal:
  - Identify incremental improvements (where, how, if possible)
  - Methodology to implement
  - Deployment plan

- Summary proposal format: open.

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